Mitacs BSI Application Template

Template version: 2025-12-03

Upon submitting your application, you are agreeing that Mitacs can post the title of your project, the public project overview, the name of the partner organization (if applicable), the name of the intern, the name of the academic supervisor, and the involved academic institution on www.mitacs.ca/en/projects and this information may be used by Mitacs to publicize its programs. Mitacs Privacy Policy can be found at www.mitacs.ca/en/privacy-policy.

Section 1: Project Overview

P-1.1 Title (255 characters max)

Note: This can be changed later

UNISON: Data-Driven Optimization of Public Safety Resource Deployment

P-1.2 Size of application

☒ Small (1-3 internships)

Expert review should take no more than 6 weeks for the majority of proposals. Please plan for additional review time should revisions be requested. Your project start date will only be confirmed once the proposal is approved, intern(s) identified, and partner organization funds received at Mitacs.

☐ Medium (6-10 internships)

Expert review should take no more than 8 weeks for the majority of proposals. Please plan for additional review time should revisions be requested. Your project start date will only be confirmed once the proposal is approved, intern(s) identified, and partner organization funds received at Mitacs.

P-1.3 Public Project Overview (100-200 words)

Using simplified language, please provide a general one-paragraph description of the proposed project including the expected benefit to the partner organization.

The UNISON project will enhance an existing public safety predictive analytics tool used by the City of Edmonton to optimize the deployment of public safety resources. The Data Science and Research team will refine existing predictive models for improved accuracy and precision in forecasting calls for service, integrate new data sources to enrich models and gain deeper insights into public safety trends, and design and prototype new data-driven features such as localized prediction capabilities and advanced visualization tools. Additionally, the team will develop robust metrics to evaluate the impact of UNISON's predictions on public safety outcomes. These

enhancements will enable more informed, proactive deployment of public safety resources across the city, leading to improved efficiency and better outcomes for the community while contributing to a scalable data science framework that supports the application's continued growth.

P-1.4 Academic discipline

Please highlight the academic discipline (1 only) that most closely aligns with the project.

- ☐ Business
- ☒ Computer Science
- ☐ Earth Science
- ☐ Engineering
- ☐ Life Science
- ☐ Mathematical Science
- ☐ Physical Science
- ☐ Social Science, Arts and Humanities

P-1.5 Project priority sectors

Please select the project's priority sectors (3 max)

- ☐ Advanced Manufacturing
- ☐ Aerospace
- ☐ Agriculture & Food
- ☐ Aquaculture & Fishing
- ☒ Artificial Intelligence
- ☐ Automotive
- ☐ Biomanufacturing
- ☐ Biotechnology
- ☐ Cannabis
- ☐ Clean Technology
- ☐ Commercial Services
- ☐ Construction
- ☐ Cyber Security
- ☐ Education
- ☐ Energy & Utilities
- ☐ Entertainment & Media
- ☐ Env. Science & Technology
- ☐ Finance & Insurance
- ☐ Forestry
- ☐ Green/Alternative Energy
- ☐ Health/Related Sci & Tech
- ☐ Indigenous Innovation
- ☐ Info & Communication Tech
- ☐ Life Sciences (not health)
- ☐ Manufacturing & Construction
- ☐ Mining
- ☐ Nanotechnology
- ☐ Natural Gas
- ☐ Natural Resources
- ☐ New & Digital Media
- ☐ Ocean Tech
- ☐ Oil & Gas
- ☐ Other
- ☐ Pharmaceuticals
- ☒ Pub. Services, Policy & Governance
- ☐ Quantum Science
- ☐ Social Innovation
- ☐ Sustainability & the Env.
- ☒ Technology
- ☐ Tourism
- ☐ Transportation
- ☐ Water

P-1.6 Keywords to identify reviewers

Please specify five to ten keywords to help identify proper reviewers and to accelerate the review process. Please provide both technically and/or conceptually related (50%) and discipline related keywords (50%).

Conceptually/technically-related keywords (50%)	Discipline-related keywords (50%)
• Predictive analytics • Resource optimization	• Data science • Machine learning

• Geospatial analysis • Data visualization • Data integration	• Public safety • Operations research • Statistical modeling

Section 2: Academic Supervisor Details

P-2.1a Academic Supervisor details

Please complete the following table for each academic supervisor included on the project (copy the table as necessary).

ACADEMIC SUPERVISOR FULL NAME	
First name:	Leia
Last name:	Minaker
Email:	lminaker@uwaterloo.ca
Re-type email:	lminaker@uwaterloo.ca
Contact phone:	519-888-4567
Extension:	45615
Academic institution:	University of Waterloo
Department:	School of Planning
Address:	200 University Avenue West
City:	Waterloo
Province:	ON
Country:	Canada
Postal code:	N2L 3G1

P-2.1b Academic Supervisor Policy on Sensitive Technology Research and Affiliations of Concern Declaration

1. Are you currently affiliated with, or in receipt of funding or in-kind support from any of the listed Named Research Organizations (NROs)?

   ☐ Yes ☒ No

   Any applicant who is currently affiliated with, or in receipt of funding or in-kind support from one or more of the institutions on the list of NROs is not eligible to participate in a Mitacs-funded project involving research that aims to advance a Sensitive Technology Research Area (STRA). Mitacs will also screen a sample of applications, selected at its sole discretion, and validate the accuracy of the completed declarations. Mitacs reserves the right to decline funding, at any point, for a project that advances a sensitive technology research area because of affiliations of concern.

P-2.1c Academic supervisor CV

The academic supervisor must upload a CV/resume for any application that will involve 6 or more internships. Please note by uploading your Resume/CV you are granting all participants in this application and Mitacs staff access to this document.

Not required for this application.

Section 3: Partner Organization Information

P-3.1 Organizational Profile

Please complete the following table for each partner organization included on the project (copy the table as necessary)

PARTNER ORGANIZATION Organization Name (Primary contact)			
Partner Contact Information			
First name:	Zehra	Last name:	Shah
Email:	zehra.shah@edmonton.ca		
Department:	Data Science and Research		
Contact phone:	780-508-9527	Extension:	
Business Information			
Legal Organization Name:	City of Edmonton		
Organization URL:	www.edmonton.ca		
Industry code (NAICS):	913: Local, municipal and regional public administration		
Legal Status: (For Canadian Organizations only)	☐ For Profit Canadian Corporation ☐ Not for Profit Canadian Corporation ☐ Crown Corporation ☒ Municipality		
Organization Size: (Number of employees in Canada)	☐ 1 – 49 ☐ 50 – 99 ☐ 100 – 499 ☐ 500 – 999 ☒ 1000+		
Address:	1 Sir Winston Churchill Square		
City:	Edmonton	Province:	AB
Country:	Canada	Postal code:	T5J 2R7
Organizational Structure and Activities			
Is the organization incorporated?	☒ Yes ☐ No	**If yes,** *please include the date (YYYY/MM/DD) of incorporation:* 1904-10-06	
Is the organization a parent company?	☐ Yes ☒ No	**If yes,** please describe (e.g., main subsidiaries, name of the parent company/holding company, city and country where the headquarters of the parent company/holding company are located): n/a	
Is the organization a subsidiary company?	☐ Yes ☒ No	**If yes,** please describe (e.g., main subsidiaries, name of the parent company/holding company, city and country where the headquarters of the parent company/holding company are located): n/a	

PARTNER ORGANIZATION Organization Name (Primary contact)		
Does the organization have an R&D department in Canada?	☐ Yes ☒ No	**If yes,** what is the exact number of research and development (R&D) staff in Canada? n/a **If no,** is R&D conducted on the organization's premises in Canada? No
Is this the first time the partner has collaborated with the academic institution?	☐ Yes ☒ No	
Funds Declaration		
Have the funds committed by the Organization been leveraged against other federal or provincial programs?	☐ Yes ☒ No	**If yes,** Please contact your Mitacs Advisor before you proceed with the application. **If yes,** please identify the Program Name(s): n/a **If yes,** please indicate the Total Program(s) Contribution: n/a
Invoicing Information		
How often will the partner organization be invoiced?	☐ Every 4 months ☐ Annually ☒ Single payment	
Net invoice term (Days >=30)	30	
Accounts payable attention to:		
First name:	Zehra	
Last name:	Shah	
Email:	zehra.shah@edmonton.ca	
Phone:	780-508-9527	
Extension:		
Please describe any applicable invoice requirements:	n/a	

Section 4: Project details

4.1 PROJECT DESCRIPTION

A-4.1a Background (What problem needs to be solved?)

Provide an overview and describe the main activities of the partner organization. Explain the innovation challenge or improvement priority that the partner needs to solve. Specify how this project will help the partner organization address those challenges in a way that goes beyond day-to-day business operations/activities. Describe what kind of expertise is required to solve the problem.

Overview of partner organization

The City of Edmonton's Data Science and Research (DSR) team acts as a central problem-solving hub, collaborating with diverse internal municipal teams by accepting projects based on a structured evaluation process. It is dedicated to leveraging data and technology to enhance municipal services and improve quality of life for residents. This team leads initiatives in open data, artificial intelligence and predictive analytics, and collaborates with various stakeholders to drive innovation and transparency. Their work includes developing predictive models for emergency events, optimizing urban planning through data-driven insights, and fostering a culture of open government through accessible data portals.

The team acts as a central problem-solving hub, collaborating with diverse internal municipal teams by accepting projects based on a structured evaluation process. This process scores requests against the City's Corporate Business Plan and their potential to generate value (make/save money, reduce risk, save operationalizing the results, projects are prioritized and initiated.

The DSR team emphasizes strong relationships, a well-defined problem, willing stakeholders, clarity on expected changes, and comfort with unpredictable results as critical for project success, all while maintaining a strong focus on data privacy and ethics.

Description of current challenge

The City of Edmonton faces the ongoing challenge of meeting its operational needs for public safety resource deployment and ultimately keeping the public safe. The City's Data Science and Research (DSR) team has developed a public safety predictive analytics tool that leverages data to inform the proactive deployment of public safety resources.

This project involves reviewing the current operational support structure to ensure optimal positioning for scaling and maintaining the application's data science capabilities. Specifically, the DSR team needs to improve the accuracy and precision of existing predictive models that forecast calls for service, integrate diverse data sources to enrich these predictions and uncover deeper insights into public safety trends, and develop advanced features—particularly localized prediction capabilities—that provide more granular, actionable intelligence for resource deployment decisions.

The DSR team successfully developed and deployed the Unison Application, a sophisticated, unified platform designed to centralize and manage all critical City of Edmonton incidents as well as forecasting for the next day where the hotspot is. This powerful system achieves comprehensive situational awareness by integrating real-time data streams from eight diverse internal and external sources, providing a single source of truth for city operations and emergency management.

The strategic roadmap for the ongoing development of the Unison Application focuses on three key priority areas to further enhance its capabilities and value:

- **Advanced Forecasting Capabilities:** The DSR team is actively working on implementing next-generation forecasting models. These new models will move beyond simple historical trends by incorporating complex spatial and temporal relationships between various city grids—such as traffic flow, utility consumption, and environmental factors—to predict future incidents with greater accuracy and lead time. This capability will significantly improve proactive resource allocation and operational planning.
- **User Experience and Operational Enhancements:** A core priority is continuous improvement based on actionable insights derived from direct user feedback. Planned enhancements include the addition of new, highly requested features focused on improving data accessibility, providing more intuitive and interactive data visualization tools, and streamlining workflows. These updates are designed to ensure the Unison platform remains an efficient and indispensable tool for all city personnel.
- **Incident Consolidation through Generative AI:** The DSR team is pioneering the implementation of Generative Artificial Intelligence (Gen AI) to revolutionize the incident consolidation process. This new Gen AI approach will precisely consolidate high-volume, related incident calls at the granular call type level. This represents a major technological upgrade from the platform's current system, which relies on a heuristic AI approach that only matches and consolidates incidents at the less detailed higher occurrence type level. The move to Gen AI will dramatically increase the precision and fidelity of incident grouping, leading to faster, more accurate analysis and response.

<u>Impact of project</u>

The Unison project is a critical application for safety deployment, involving collaboration between internal and external teams. Enhancing its capabilities by providing more accurate forecasting—through the consideration of location relationships and environmental factors—will not only improve immediate predictions but also allow for multi-day forecasting, facilitating better planning for upcoming weeks. Furthermore, improvements to the user interface will enable users to define their areas of interest, filtering the volume of information to present exactly what is relevant to their needs. Providing forecasts at a more granular, call-type level will also allow teams to focus on more specific and detailed information. In addition, by establishing clear metrics to measure UNISON's impact, the City will be able to quantify how the tool is contributing to making communities safer and continuously improving it based on real results. This goes beyond day-to-day operations by fundamentally strengthening the City's ability to proactively protect residents rather than responding to incidents after they occur.

Expertise required to solve the problem
Specialized expertise required to develop this solution includes (but is not limited to):

- **Programming and Databases:** Demonstrated experience with programming languages and relational databases, particularly SQL.
- **Data Analysis and Visualization:** Proficiency in data analysis and visualization tools (e.g., R or Python).
- **Time Series Forecasting:** Practical experience with Time Series Forecasting methods.
- **Full-stack Development:** Experience with full-stack development, specifically using React and Python.

A-4.1b Objectives and Approach (How will the problem be solved?)

State the objectives for this project and explain how you plan to accomplish them. Enough detail should be provided so that it is clear whether the described approach is appropriate to achieve each objective. For projects with more than one intern, clearly explain which objective(s) each intern will work on.

The Unison project's core objectives are twofold:

- To develop a robust forecasting model for the strategic deployment of safety resources.
- To enhance system functionality to allow users to conduct customized data analysis.

These objectives will be achieved through a data-driven, iterative methodology. The initial steps will involve upgrading the user interface to align with modern frameworks and best practices. Subsequently, research will focus on the optimal way to integrate environmental factors and locational relationships into time series forecasting. This research and implementation will follow the implementation of a Generative AI model for consolidating call types.

A-4.1c Timeline (Who will solve the problem and how much effort will be required?)

*Insert a timeline showing **which task(s)** will be performed **when**, and by **which intern** in order to achieve the objectives stated above (consider breaking this down by week for shorter projects or by month for longer projects via a Gantt chart). Ensure there are no gaps in the timeline and that the effort required to complete each task matches the amount of time allocated for each intern.*

The project will proceed as follows:

TASK	PROJECT MONTHS								RESPONSIBLE
	1	2	3	4	5	6	7	8	

Incorporate new data into Unison, such as problem properties and employee safety information.									Mitacs Intern
• Refine the existing data source pulling process. • Research and implement the incorporation of location and environmental factors into forecasting models.									Mitacs Intern
• Utilize Generative AI for the consolidation of call type data • Develop functionality for forecasting various call types.									Mitacs Intern
Create customized daily reports for users.									Mitacs Intern
Migrate the current workflow to a cloud-based environment.									Mitacs Intern
Modernize the current front-end framework by transitioning to React.									Mitacs Intern
Enhance the User Interface (UI) by implementing requested user features.									Mitacs Intern
Develop comprehensive documentation for all new features and improvements.									Mitacs Intern

A-4.1d Deliverables (What does success look like?)

List the deliverables or tangible results *you expect from the project and identify which intern is responsible for it. For example, will you develop a new patent/prototype/product/service, business model, business process, or will you access new markets? Will the intern be expected to produce a report, evaluation, presentation, etc.?*

Deliverables for this project include (but are not limited to):

- **Enhancing existing products or services:** Once the predictive models have been refined and new data sources have been integrated, the DSR team will work on upgrading UNISON's user interface to align with modern frameworks and best practices and extend system functionality to provide customizable data analysis at the neighborhood level, enabling more precise resource deployment decisions across specific areas of the city.

- **Comprehensive reports and presentations for the partner organization:** These will document methodologies, findings, and recommendations related to predictive model enhancements, data integration strategies, new feature development, and performance evaluation metrics while effectively communicating technical development, analytical results, and practical implications for proactive public safety resource deployment and improved community safety outcomes.

A-4.2 INTERNSHIP BENEFITS

A-4.2a Benefits and Impacts to Canada

If this project potentially impacts broader challenges that society or the industry faces, please describe how the project will help address these challenges. Why are these contributions important?

The UNISON project addresses broader challenges that many cities and public safety organizations face in efficiently allocating limited resources to protect their communities. As municipalities across North America grapple with budget constraints, increasing service demands, and the need to demonstrate measurable outcomes, data-driven approaches to public safety resource deployment offer a path forward. By developing enhanced predictive models, integrating diverse data sources, and establishing robust performance metrics, this project contributes to the growing field of smart city analytics and evidence-based public safety practices. The focus on building a scalable data science framework and exploring advanced methodologies like MLOps and causal inference creates knowledge and approaches that can be adapted by other jurisdictions facing similar challenges.

Ultimately, this work demonstrates how cities can leverage data science to move from reactive to proactive service delivery, improving both the efficiency of resource use and the safety outcomes for residents—challenges that are increasingly urgent for communities everywhere.

A-4.2b Benefits to the Intern(s)

Explain how participating in this project aligns with the academic studies of the intern(s). How is participating in the project expected to benefit their future career(s)? For example, will they expand their professional network, gain knowledge in a business context, or develop skills in problem-solving, communication, project management, creative thinking, etc.?

By participating in this project, the intern will:

- Apply their academic expertise to real-world problems by using the specific skills mentioned in Q 2.1.a to refine predictive models, integrate and analyze diverse data sources, develop new data-driven features, and evaluate the performance and impact of public safety analytics tools.
- Use and develop creative problem-solving skills by thinking on their feet and solving problems in real time

- Foster management, leadership, and communication abilities by working with an experienced team and ensuring the end product delivers accurate predictions, actionable insights, and measurable improvements in public safety resource deployment.
- Gain valuable experience in applying data science methodologies to address complex societal challenges while contributing to innovations that directly improve community safety and well-being.
- An intern leading a data science project for the City of Edmonton will gain hands-on experience applying theoretical concepts—such as machine learning, statistical modeling, and programming—to complex municipal challenges. This opportunity will expand their professional network within a leading municipal data science team, deepen understanding of public sector operations, and enhance problem-solving skills by addressing novel issues. Additionally, the experience will build skills in project management, stakeholder communication, ethical data practices, and the ability to operationalize data science solutions from concept to deployment.

A-2.3 INTERNSHIP ENVIRONMENT

A-2.3a Description of the Academic Internship Environment

a. *Please describe the nature of academic supervision during the internship period. Explain the **type and frequency of interactions** (e.g., meetings, phone calls, etc.) between the intern and academic supervisor.*
b. *Will the project make use of the resources and/or facilities of the academic institution? If so, please describe.*

Academic internship Environment

The Future Cities Institute (FCI), founded by CAIVAN is committed to fostering healthy, prosperous cities and communities in Canada and beyond through interdisciplinary partnerships, action-oriented research, and data-informed decision-making. The Data Science and Research Team at the City of Edmonton is one of FCI's core partners due to their focus on developing data-driven solutions to municipal challenges and their commitment to the democratization of tools and resources.

Nature of academic supervision

The intern will meet virtually with Dr. Leia Minaker (Director, Future Cities Institute, Faculty of Environment) at least once a month to ensure ongoing alignment with the goals and objectives of the Future Cities Institute and to discuss progress towards knowledge mobilization outcomes.

Use of UWaterloo resources and/or facilities

The initial development, implementation and evaluation of the solution will use very few UWaterloo resources and/or facilities since the intern is physically located in Edmonton. However, the FCI staff team (primarily Research Coordinator and Communications Officer) will be engaged throughout the internship to support knowledge mobilization and/or communication activities.

A-4.3b Description of the Partner Internship Environment

a. *Please state the nature of the partner interaction:* ☒ *Onsite* ☐ *Virtual* ☐ *Hybrid*

b. *Describe the working environment at the partner organization and the type and frequency of interactions between the intern(s) and employees. For example, what resources, facilities/equipment, and specialized training will the intern(s) have access to? If the intern(s) will work virtually, how will they be exposed to the activities and culture of the partner organization?*

Partner Internship Environment

The intern will immerse themselves in the Data Science and Research team's dynamic, pragmatic, and highly collaborative environment at the City of Edmonton. Interactions will be frequent and direct, involving continuous engagement with senior data scientists for mentorship and active collaboration with partner teams across various city departments, from project intake through operationalization, to define problems and ensure solutions meet real-world needs.

Access to resources, facilities/equipment and specialized training

For resources, the intern will access state-of-the-art centralized infrastructure, including Google Cloud Platform services (BigQuery, Dataflow, Dataproc, GPU support) and GitLab for project management and version control. Specialized training will be hands-on, provided through direct guidance from experienced public sector data scientists, ensuring practical skill development.

Organizational culture

Cultural exposure will stem from direct team integration, participating in the team's problem-solving ethos, and gaining deep institutional knowledge by observing and contributing to the operational realities and diverse activities of the partner organizations they support.

A-4.4 PAST/OTHER MITACS PROJECTS

Is this application related to, or a continuation of, any previous or current Mitacs projects?

☐ *Yes*

☒ *No*

*If **Yes**, provide specifics about the relationship(s), including relevant project IT#(s). If the current project is a continuation of prior work, briefly describe how it meaningfully furthers what has been previously achieved.*

n/a

A-4.5 ENGAGEMENT WITH INDIGENOUS COMMUNITIES OR PARTNERS (IF APPLICABLE)

Projects that involve or impact Indigenous communities must comply with the Mitacs Indigenous Research Policy. Describe (1) Indigenous community support for the project, and their role in shaping its objectives/approach, (2) plans for Indigenous community access, use, and governance of resulting knowledge/data, and (3) the team background in Indigenous research, including planned training/mentorship the intern(s) will receive to address deficits in experience.

You may also submit 1-2 letter(s) of support from Indigenous Elders who are members of the partner community/communities and possess the authority to speak on community interests.

n/a

A-4.7 INTELLECTUAL PROPERTY (IF APPLICABLE)

*Explain how this project supports the **creation and/or ownership of IP** for the partner organization. If the project provides **education or training about IP** for any of the participants, explain here.*

Any intellectual property created by a Placement Student during their participation in the Program or performance of the Work will be the exclusive property of the City of Edmonton (the "Intellectual Property).

The City of Edmonton may, at its discretion, provide advance written consent to a Placement Student for their use of the Intellectual Property for non-commercial purposes.

The City of Edmonton also agrees to facilitate the reasonable use of the Intellectual Property by the University of Waterloo for academic purposes related to the administration of a Placement Student's education, where applicable.

Subject to the protection of any confidential information of the City of Edmonton or information otherwise protected under applicable legislation, including the Freedom of Information and Protection of Privacy Act (RSA 2000, c F-25), as amended, this reasonable use by the University of Waterloo will include posting each Placement Student's final reports and presentations related to the Work on the University's website.

A-2.7 REFERENCES / SOURCES (IF APPLICABLE)

List any references you have cited in your proposal.

n/a

Section 5: Internship Budget and Schedule

P-5.1 SUMMARY OF INTERNSHIPS

Please select only one type of internship option per application from the list below. If you want to apply for internships of a different type, you will need to create a separate application. Each of your applications can include up to 10 internships of the same type. If you select an internship option with project costs, you must include a description of how these fundings will be used in your project proposal.

☐ **Business Strategy Internship (BSI) Standard Funding:**
Each BSI internship unit is valued at $10,000 and the partner organization's financial contribution starts at $5,000 per unit. Each intern receives a minimum of $10,000 stipend per unit

☒ **Business Strategy Internship (BSI) Standard Funding plus Project Costs:**
Each BSI internship unit is valued at $15,000 and the partner organization's financial contribution starts at $7,500 per unit. Each intern receives a minimum $10,000 stipend per unit. The additional $5,000 in funds can be used to pay for eligible project costs or can be used to top-up the intern stipend to $15,000.

☐ **Business Strategy Internship (BSI) - Indigenous Pathways Reduced Cost 10K:**
Projects with Indigenous participants are eligible for a reduced partner contribution. Each BSI internship unit is valued at $10,000 and the partner organization's financial contribution starts at $2,500 per unit. Each intern receives a minimum $10,000 stipend per unit.

☐ **Business Strategy Internship (BSI) - Indigenous Pathways Reduced Cost 15K:**
Projects with Indigenous participants are eligible for a reduced partner contribution. Each BSI internship unit is valued at $15,000 and the partner organization's financial contribution starts at $3,750 per unit. Each intern receives a minimum $10,000 stipend per unit. The additional $5,000 in funds can be used to pay for eligible project costs or can be used to top-up the intern stipend to $15,000.

P-5.2 INTERNSHIP DETAILS

P-5.2a Intern Name

** NOTE: An intern can be named (if known) or TBD*

[INSERT TEXT]

P-5.2b Degree Program During Internship

NOTE: A recent graduate is an individual who, at the start of the internship, graduated less than 2 years ago. "Other University Program" should be used for an individual who is enrolled at a

Canadian university in a non-degree program (e.g., diploma, certificate). "Recent Graduate (Other University Program)" should be used for an individual who, at the start of the internship, graduated less than 2 years ago from a non-degree program (e.g., diploma, certificate).

- ☐ College
- ☐ Bachelors
- ☐ Masters
- ☐ PhD
- ☐ PDF
- ☐ Recent Grad – College
- ☐ Recent Grad - Bachelors

P-5.2c Internship timing

a. *Start date:*
b. *Duration (months):*
c. *# of Consecutive Internships With Same Stipend (see * below):*
d. *End date: This auto-generates in the system based on the start date and the duration of the internship*

> * NOTE: 'Consecutive internships' refers to sequential internships performed by a specific intern without breaks (gaps) in between each. To create a break between internships, create a new row for that specific intern. To enable each internship for a given intern to have a different stipend value from the other internships for the same intern, please create a new row for that specific intern. 'Same stipend' refers to every internship for that specific intern being assigned the same stipend value. Tip: Use the Duplicate function in the Actions menu to copy internships that are similar then make simple edits (such as modifying the stipend value, internship length, or add a new intern with the same stipend, duration and sequence as previous additions) as outlined above.

P-5.3 ACADEMIC SUPERVISORS

P-5.3a Participant details

This information will be auto-generated from the "Participants" tab.

P-5.3b Partner organizations

The total minimum required organization contributions are listed in section P-3.1 above. This contribution can be distributed across multiple organizations.

a. *Please note that the financial contribution of organizations is subject to applicable Goods and Services Tax (GST), Harmonized Sales Tax (HST) and/or Quebec Sales Tax (QST) (Collectively the value added tax or VAT).*

b. *Partner organizations can contribute additional funding to the project but Mitacs will not provide any additional matching funding outside of the standard awards.*
c. *Additional partner contributions are only allowed with academic institution approval.*
d. *Any additional partner funding must be shown in the proposal/budget and the anticipated use of these funds must be explained so that the academic institution can clearly see and approve the planned budget when they sign-off on the proposal (e.g. will the extra funds be used for stipend top-up, specialized equipment, etc.?).*
e. *Any additional funds associated with a specific internship must be received by Mitacs at the same time as the normal partner contribution.*
f. *The funds will be added to the award and forwarded to the academic institution to be paid out to the intern.*

PARTNER NAME	IS A PARTICIPANT OF THIS INTERNSHIP (See * below)	AMOUNT PER INTERNSHIP (See ** below)
Partner A	*Auto-generated based on "Participants" tab*	*$7,500*

** Make sure each participant has been entered in the "Participants" tab.*
*** Note this amount does not include the tax component*

P-5.4c Stipends and Project Expenses:

a. *Partner organizations can contribute additional funding to the project but Mitacs will not provide any additional matching funding outside of the standard awards.*
b. *Additional partner contributions are only allowed with academic institution approval.*
c. *Any additional partner funding must be shown in the proposal/budget and the anticipated use of these funds must also be shown so that the academic institution can clearly see and approve the planned budget when they sign-off on the proposal.*
d. *Any additional funding associated with a specific internship must be received by Mitacs at the same time as the normal partner contribution.*
e. *The funds will be added to the award and forwarded to the academic institution to be paid out to the intern.*
f. *A minimum of $10,000 per internship must be use for intern stipend. Any additional funding the partner organization provides will be allocated to the intern stipend amount.*

BUDGET ITEM	AMOUNT
Stipend (must be at least $10,000)	
Project Expenses (usually $5,000)	
Total	

P-5.4d Details of project expenses.

If your internship includes project expenses, you must enter details of these project expenses in the space provided. For example, if you have two four-month internships listed above, please include what expenses will be required for each of these two units.

The project expenses for the XX IU(s) included in this application will be used to top-up the intern's stipend.

Section 6: Declarations

6.1 DECLARATIONS FROM PORTAL

P-6.1 Answer the following questions:

Declarations from the Portal		
Will the intern conduct any of the proposed project activities outside of the academic institution or partner premises?	☐ Yes ☒ No	
Does the project involve:		
Human participant whose data, or responses to interventions, stimuli, or questions by the intern, are relevant to completing the work?	☐ Yes ☒ No	***If yes,*** *was checked for any of these questions, applicants must contact the ethics offices at all participating academic institutions to determine whether ethics clearance is required. This includes primary/secondary use of human data/biological materials owned by the partner.*
Secondary use of human data or health information (even if anonymized)?	☐ Yes ☒ No	
Human biological materials, as well as human embryos, fetuses, fetal tissue, reproductive materials, and stem cells? This applies to materials derived from living and deceased individuals	☐ Yes ☒ No	
Secondary use of biological materials (even if anonymized)?	☐ Yes ☒ No	
Will the proposed project require the use of laboratory animals, and/or potentially impact the well-being of wild/domesticated animals?	☐ Yes ☒ No	***If yes,*** *applicants must contact the animal care committees at all participating academic institutions to determine whether any certifications/authorizations are required.*
Will the proposed project require the interns to handle or be exposed to biohazards?	☐ Yes ☒ No	***If yes,*** *please check all that apply:*
		Biohazards (e.g., viruses, bacteria, fungi, parasites, toxins, prions, zoonotic pathogens, recombinant DNA, genetically modified organisms, viral vectors, synthetic organisms, cell lines/cultures) — ☐ Yes ☐ No
		Radioactive materials — ☐ Yes ☐ No
		Restricted substances (e.g., cannabis) — ☐ Yes ☐ No
		Other — ☐ Yes ☐ No
		If one or more options were checked, any necessary documentation must be obtained in accordance with all participating academic institutions' policies and maintained throughout the duration of the research project.

Declarations from the Portal		
Will the project involve the use of personal data or large datasets that could be considered sensitive (for additional information, see List 2 of Annex A of the National Security Guidelines for Research Partnerships)?	☐ Yes ☒ No	***If yes,*** *please describe:* RESPONSE
Will the project involve research related to critical minerals and/or one of the critical infrastructure sectors?	☐ Yes ☐ No	***If yes,*** *please describe:* RESPONSE
Will the project involve research related to goods or technology that are included on the Export Control List (ECL) of the Export and Import Permits Act (EIPA)?	☐ Yes ☒ No	***If yes,*** *please describe:* RESPONSE
Will background intellectual property (IP) be transferred from Canadian institution(s) to the partner organization(s)?	☐ Yes ☒ No	***If yes,*** *please describe:* RESPONSE
Will intellectual property (IP) arising from the project be owned by, assigned to, or licensed to the partner organization(s)?	☐ Yes ☒ No	***If yes,*** *please describe:* RESPONSE
Does the project aim to advance any of the listed Sensitive Technology Research Areas (STRAs)?	☐ Yes ☒ No	***If yes,*** *please describe:* RESPONSE If yes, applicants should consider any potential research security risks associated with the nature of the proposed activities. Applicants should also consult the policies, guidelines, and requirements of their participating Canadian academic institution(s) to determine any appropriate research security risk mitigation measures based on federal and provincial government guidelines/policy statements, and to identify any necessary actions to ensure compliance with relevant government regulations and legislation (e.g., import/export laws)

6.2 DECLARATIONS FROM APPLICATION FORM

A6.2a Conflict of interest

*Mitacs **will not approve a BSI application** if the intern listed on the proposal has a position of ownership, employment, or influence over the daily operations of the partner organization. Interns with close family or intimate relationships with academic supervisors or employees of the partner organization are also ineligible. Any other real or perceived conflict of interest (COI) - for example, **past intern employment at the partner organization** - **must** be disclosed to Mitacs during the application process using the Intern Eligibility and Conflict of Interest*

Declaration Form. Academic supervisors must disclose any COIs through the Academic Institution Acknowledgement of COI(s) Form, and submit this documentation on behalf of their post-secondary institution when applying through the Registration and Application Portal (RAP).

Mitacs will first determine if BSI program rules are satisfied, and if so Mitacs will then review the COI to determine whether mitigations can be enacted to manage the COI. If a COI does exist, we suggest that you contact your Mitacs Advisor prior to starting an application to ensure that the participants in this project are eligible for the program. Please view the Mitacs Conflict of Interest Policy for more information.

PLEASE NOTE: Failure to disclose a COI will:

- ***Delay the project evaluation process.***
- ***Require submission of a COI mitigation approach*** *satisfactory to Mitacs before project work initiates or continues.*
- ***Risk forced withdrawal from the BSI program****, should a satisfactory mitigation plan not be reached.*

Have any applicants declared a COI via the appropriate documentation?

☐ Yes, an intern
☐ Yes, an academic supervisor
☒ No

A-6.2b Indigenous engagement

Please indicate if any of the below statements are true.

☐ One or more named interns self-declares as an Indigenous person
☐ The partner organization is a for-profit organization with self-identifying Indigenous persons who hold 50% or greater ownership shares
☐ The partner organization is a not-for-profit with board membership consisting of 50% of greater self-identifying Indigenous persons
☒ The partner organization is a not-for-profit whose core mandate includes Indigenous community impact or serving Indigenous communities

** If any of the above statements are true, your organization may be eligible for enhanced leveraging via the Mitacs Indigenous Pathways Initiative. Please check with your Mitacs Advisor for more information.*

A-6.2c Type of innovation

Please select all the types of innovation you will significantly impact through your project. Refer to the following definition of innovation for further information: Defining innovation - OECD.

☒ Product innovation
☒ Process innovation
☐ Marketing innovation
☒ Organizational innovation